

SENTENCE MEANING

Meaning Representation

A meaning of a linguistic structure can be captured in a formal structure called **meaning representation**.

The framework that specify the syntax and semantic of this representation called **meaning representation languages**.

Need for a representation that can bridge the gap from linguistic inputs to the non-linguistic knowledge of the world, to perform tasks that involving the meaning of linguistic inputs

Meaning Representation

Following language tasks require some form of semantic processing of natural language:

- Answering essay questions on an exam
- Deciding what to order at a restaurant by reading a menu
- Learning to use a new piece of software by reading the manual
- Realizing that you've been insulted
- Following recipes

Linguistic elements

Knowledge of the world

Programming Analogy

In programming, we often talk about a function and its arguments

- `append(item, list) → list`
 - e.g. `append(1, [3,2]) → [3,2,1]`
 - Arguments are `(1, [3,2])`, function is `append`, result is `[3,2,1]`
- Arguments can have types to which they are restricted
 - `append(1, 6)` results in an error, since 6 is not a list
- Note that Boolean functions can be interpreted as relations
 - `append(A,B)` is true if A is an (item, list) pair and B is the list with that item appended to it
 - `append` is a relation and the pair `<(1,[3,2]), [3,2,1]>` is a member of that relation

Meaning, Arguments, and Predicates

- When we represent meaning, we similarly distinguish predicates and arguments
 - **Restaurant** (Cristinas)
 - **Likes** (Philip, Cristinas)
 - **Noisy** (Cristinas)
 - **Location** (Philip, Cristinas)

*Philip likes Cristina's that is a noisy Italian restaurant in Silver Spring, Maryland.

Meaning, Arguments, and Predicates (1/2)

Procedural semantics

- Meaning = Procedure for making something **true**
 - `[[ Location(Philip, Cristinas) ]]` =
 - Philip gets in car
 - Philip looks up Cristinas on Google Maps
 - Philip starts car
 - Philip drives following directions to Cristinas

Procedural Semantics

- Meaning = Procedure for verifying that something is **true**
 - [[Location(Philip, Cristinas)]] =
 - Look up Philip's iPhone GPS location x
 - Use Google Maps to get Cristinas GPS location y
 - Calculate distance between x and y
 - If distance is less than 10 yards, return TRUE, else return FALSE

Meaning, Arguments, and Predicates (2/2)

- Ex. Winograd 1972, SHRDLU
 - Uses inference to figure out what world has to be true to make something true, then make those things true
 - Good illustration of the study of natural language understanding using “microworlds”

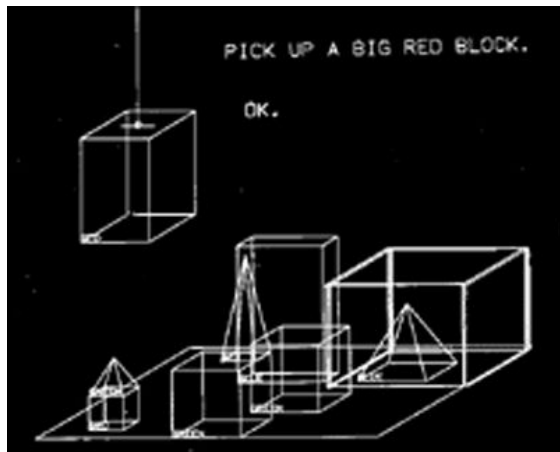


Image source:
commons.wiki
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Person: Pick up a big red block.

Computer: OK.

Person: Grasp the pyramid.

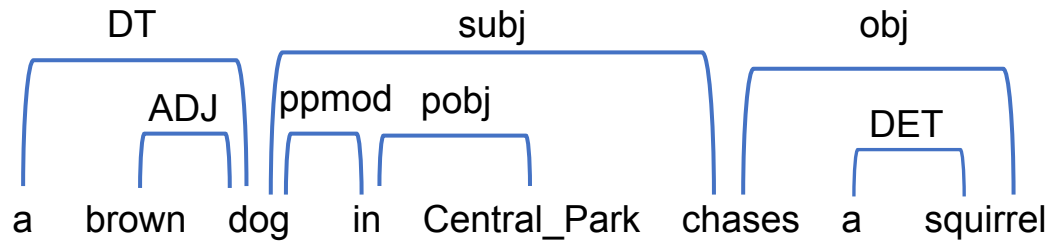
Computer: I DON'T UNDERSTAND WHICH PYRAMID YOU MEAN.

Person (changing their mind): Find a block which is taller than the one you are holding and put it into the box.

Computer: BY "IT", I ASSUME YOU MEAN THE BLOCK WHICH IS TALLER THAN THE ONE I AM HOLDING.

Computer: OK.

Model-theoretic Semantics (1/2)



Model (M)

Domain $\mathcal{D} = \{a, e, h\}$

Dog a

Squirrel e

Central_Park h

Properties

Brown $\text{Brown} = \{a\}$

– brown dog

Relations

Chases m $\text{Chases} = \{ \langle a, e \rangle \}$

– brown dog chases a squirrel

Modifiers of verb

Location $\text{Loc} = \{ \langle m, h \rangle \}$

– *chases in Central_park*

Meaning Representation (R)

$\wedge \text{chase}(m)$

$\wedge \text{subject}(m, a)$

$\wedge \text{dog}(d)$

$\wedge \text{brown}(d)$

$\wedge \text{loc}(m, h)$

$\wedge \text{squirrel}(e)$

$\wedge \text{object}(m, e)$

$\wedge \text{central_park}(h)$

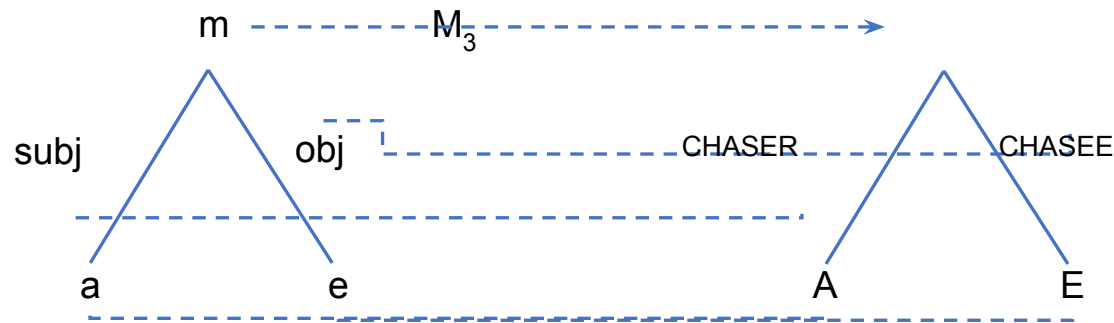
Meaning as a homomorphism

- Assume there is some model that is an abstraction of the world
- The meaning of a sentence expressed in a **meaning representation language** is a **truth value (true/false)** determined by mapping from the meaning representation to the model.

Model-theoretic Semantics (2/2)

Is the sentence true?

Is there a mapping from expression R to the model M that preserves the relationships?



Most Current Semantics Work in Computational Linguistics

- Designing meaning representation languages
- Getting from text to meaning representations (semantic analysis)
 - Traditionally via syntactic analysis
 - More recently using deep learning to go straight from language to meaning representations
- Doing things with semantic representations, especially inference

Designing Meaning Representation Languages (1/2)

- Key emphasis on semantic roles
- Predicate-argument structure – typically events and participants in that event

Leo the lion ate Remy on Tuesday

arg0

arg1

argM-TMP

Remy was eaten by Leo the lion on Tuesday

arg1

arg0

argM-TMP

Rel: verb

Argument of verb: arg0, arg1, ...

Modifiers of verb: manner (MNR), location (LOC), temporal (TMP)

Designing Meaning Representation Languages (2/2)

Proposition Bank (PropBank) is a corpus annotated with semantic roles for predicates using a linguistically motivated design of the roles

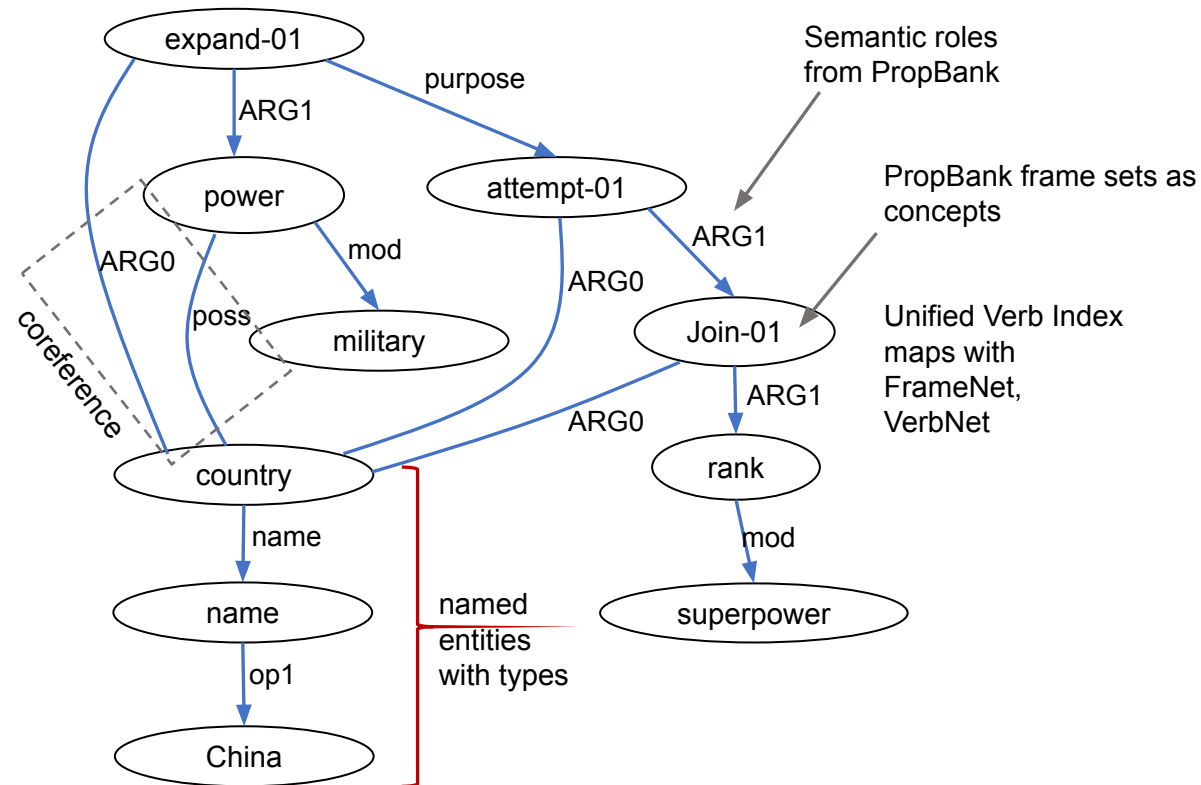
- arg0: agent, doer
- arg1: theme, done-to
- arg2: instrument (e.g., with a fork)
- ...

FrameNet is a lexical database/corpus annotation project that captures conceptual structure for an **event** with roles and associated roles

- Frame: Commerce-Goods-Transfer
(Lexical units: buy, sell, purchase)
- Core elements:
Buyer, Seller, Goods
- Non-Core elements:
Place, Purpose

Abstract Meaning Representation (1/3)

China is expanding its military power to attempt to join the ranks of the superpowers



Abstract Meaning Representation (2/3)

- Constraints on arguments of predicates
 - Similar idea to argument types in programming languages
 - Analogy: in `append(x, L)`, `L` must be a list
- Traditional approach: Boolean restrictions
 - `devour(x, y)`
 - `X` should be an animal, `y` must be edible
 - Problem: too strict
 - Jay devoured The Hobbit
 - This class is devouring my spirit

Abstract Meaning Representation (3/3)

- Information-theoretic approach (Resnik 1993, 1996)
 - Consider C to be a semantic category, e.g. in WordNet
 - Estimate $\Pr(C|V=\text{devour})$; compare to $\Pr(C)$
 - Compute KL-divergence between $\Pr(C|V=v)$ and $\Pr(C)$: how well does $\Pr(C)$ estimate $\Pr(C|V=v)$?
 - This “selectional association” between verb v and argument category C characterizes degree of constraint
 - High when C is a good semantic fit for that argument of v .

Semantic Analysis

- **SRL—semantic role labeling** - is one crucial piece of the puzzle
- Can be interpreted as a classification problem
 - [The NY Times] issued [a special solution] [yesterday]
 - Is [The NY Times] an argument? (Boolean label)
 - What is its role? (arg0, arg1, etc.) (Choice among multiple labels)
- Can also be treated as a sequence labeling problem
 - Directly analogous to POS tagging, with begins/inside/outside (B,I,O) labels
 - [The NY Times] issued [a special solution] [yesterday]
B-arg0 I-arg0 I-arg0 O B-arg1 I-arg1 I-arg1 B-argM-TMP

Compositional Semantic Analysis

- Lambda notation: a function that takes argument x and yields $P(x)$

$\lambda x \cdot P(x) \Rightarrow$ Greek symbol λ , one or more variable x , and First Order Logic formula

- Lambda-reduction: *ability to apply them to logical terms to yield new FOL; λ - reduction*

$\lambda x \cdot P(x)(A)$
 $P(A)$

Example: Using λ -expression as body of another:

$\lambda x.\lambda y.Near(x,y)$
 $\lambda x.\lambda y.Near(x,y)(Bacaro)$
 $\lambda y.Near(y)(Bacaro)(Centro)$
 $Near(Bacaro, Centro)$

- Just like in programming languages:

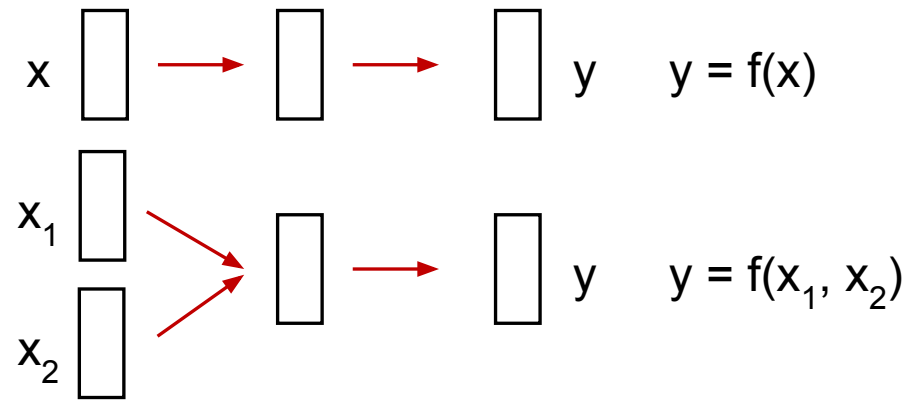
• `map (lambda x: x*2, [1,2,3])` $\square$ `[2,4,6]`

Doing Things with Representations

- Inference! Typically against a knowledge-base or knowledge graph
- Ex. Question-answering:
 - $\text{CanPatent}(X, Y) \quad \text{Inventor}(X, Y) \wedge \text{Person}(X)$
 - $\text{CanPatent}(X, Y) \quad \square \quad (\text{Organization}(X) \wedge \exists Y (\text{Person}(Y) \wedge \text{WorksFor}(Y, X)))$
- Wide range of logical formalisms supporting inference

Neural Sequence Models and Sentence Representation

- Non-sequential models

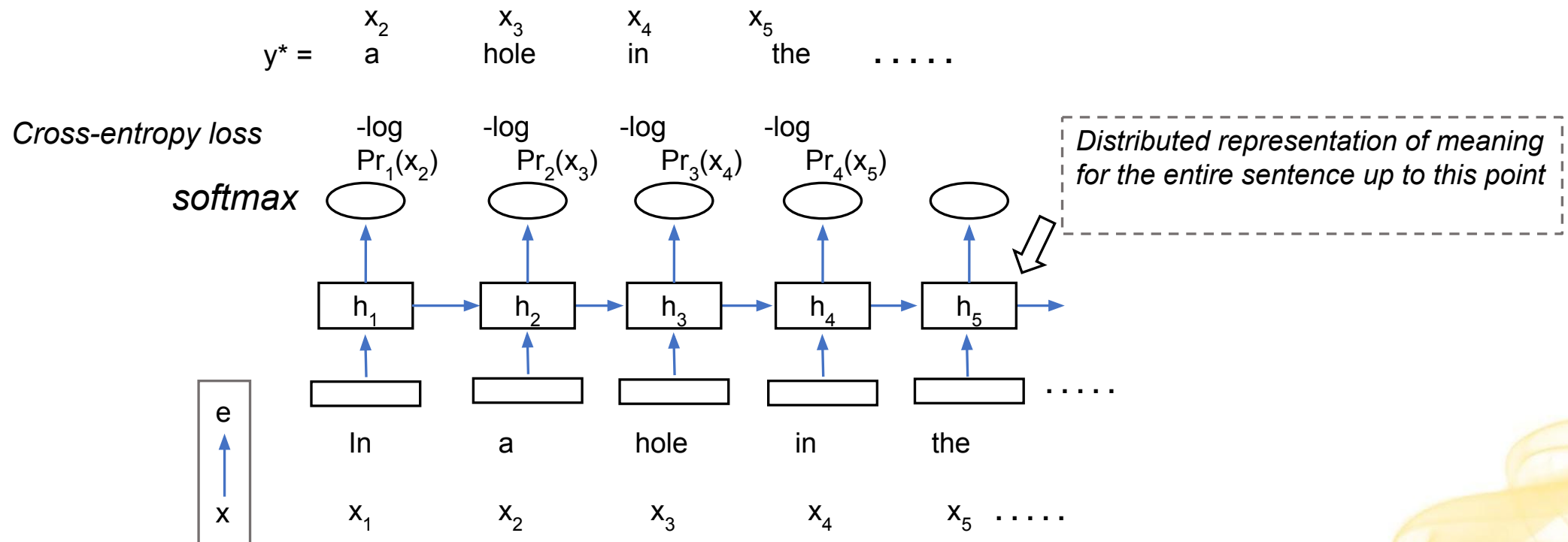


- Keep adding $x_3, x_4, \dots$?
 - Sentences grow unboundedly
 - Missing generalizations

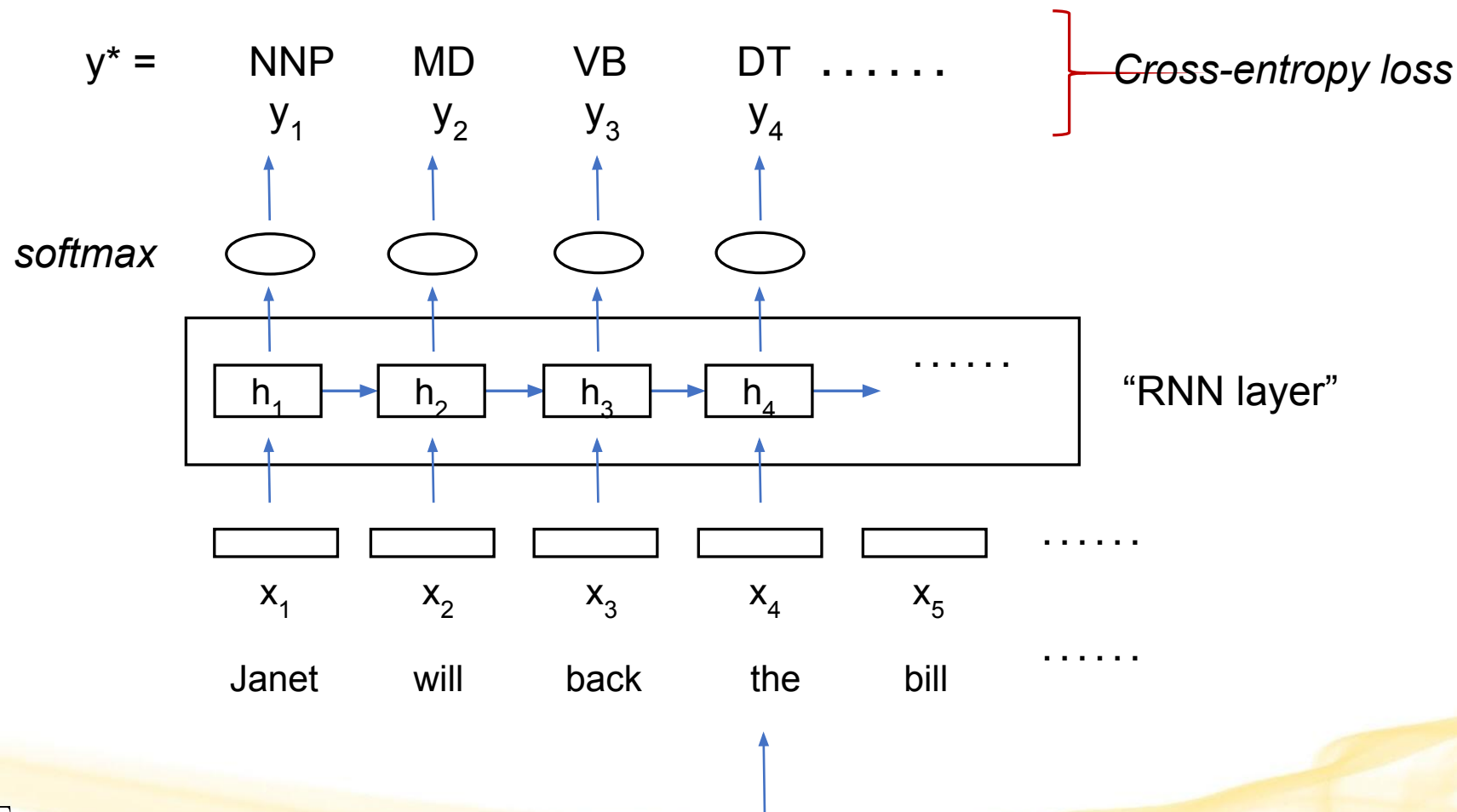
Training Recurrent Neural Networks (1/2)

- Back-propagation in time
- “Unrolls” to create a very deep network
- Input: typically symbols x are mapped immediately to pre-trained embeddings e

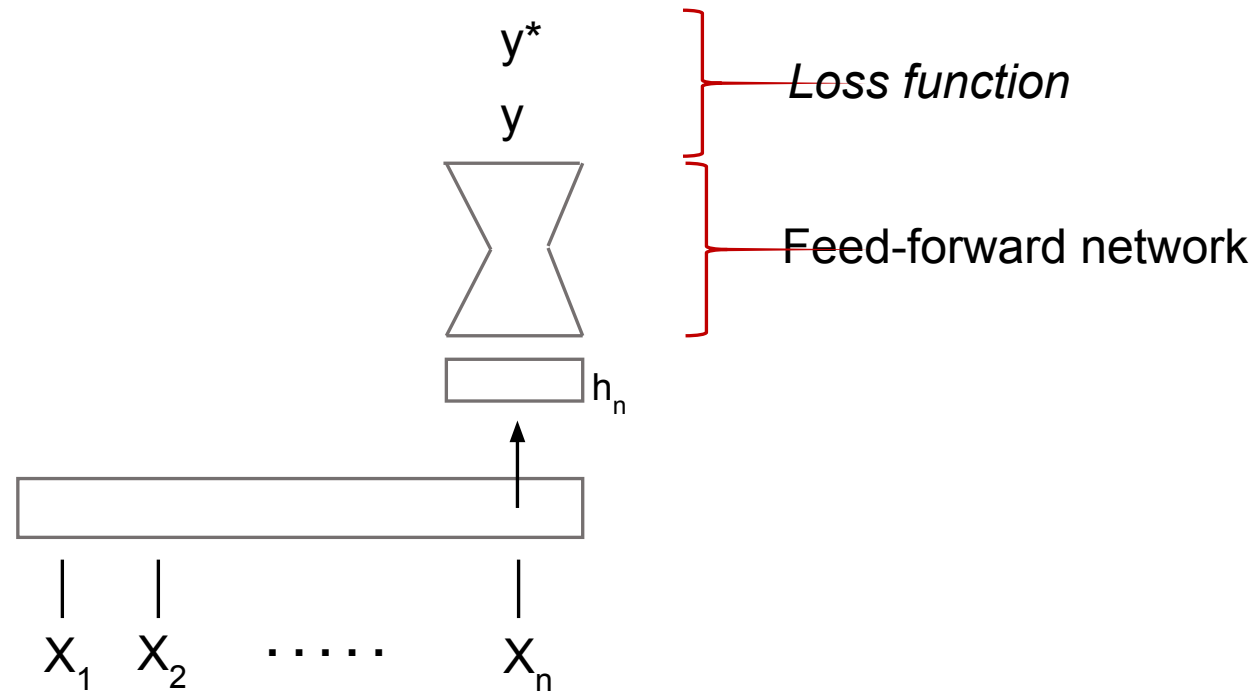
Training Recurrent Neural Networks (2/2)



Part-of-Speech Tagging with RNNs



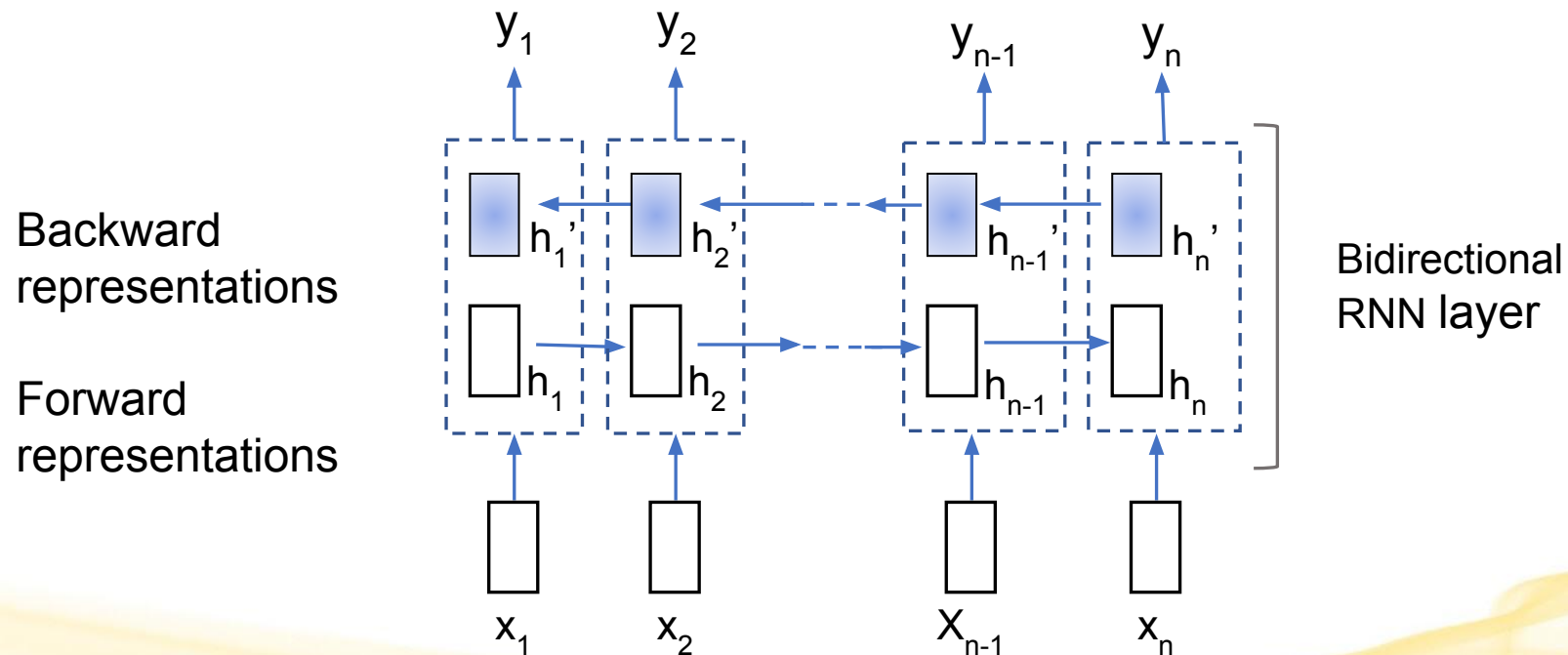
Sequence Classification with RNNs



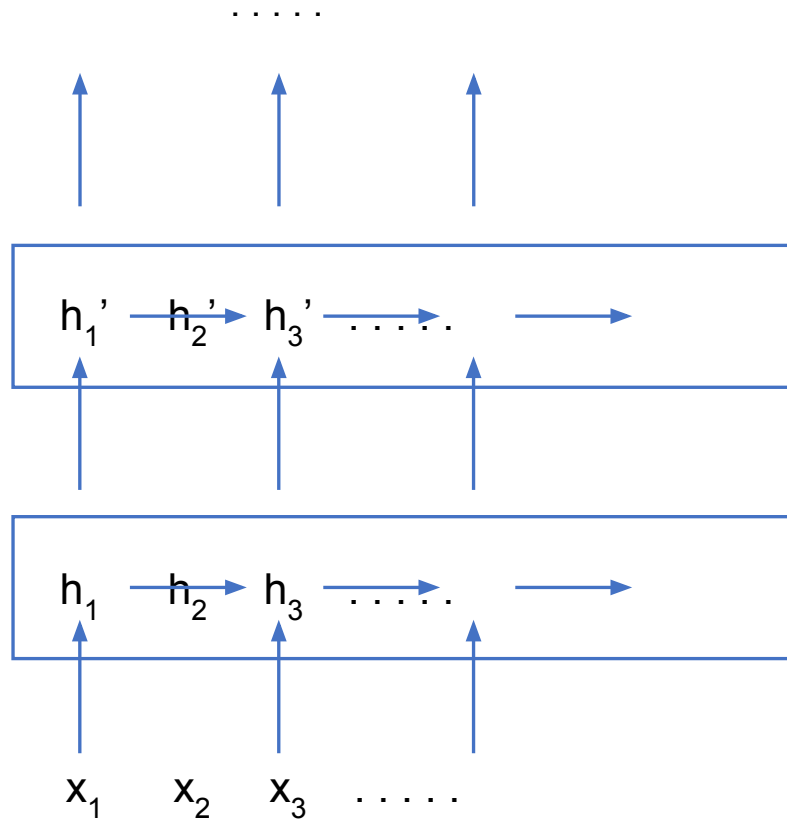
Note: back-propagation from y all the way back through the network **requires** the ability to train **deep** networks

Bi-directional RNNs

- Backward representations, forward representations
- Allows hidden representations of a token to take into account both preceding and following context
- Assumes entire string is available at once

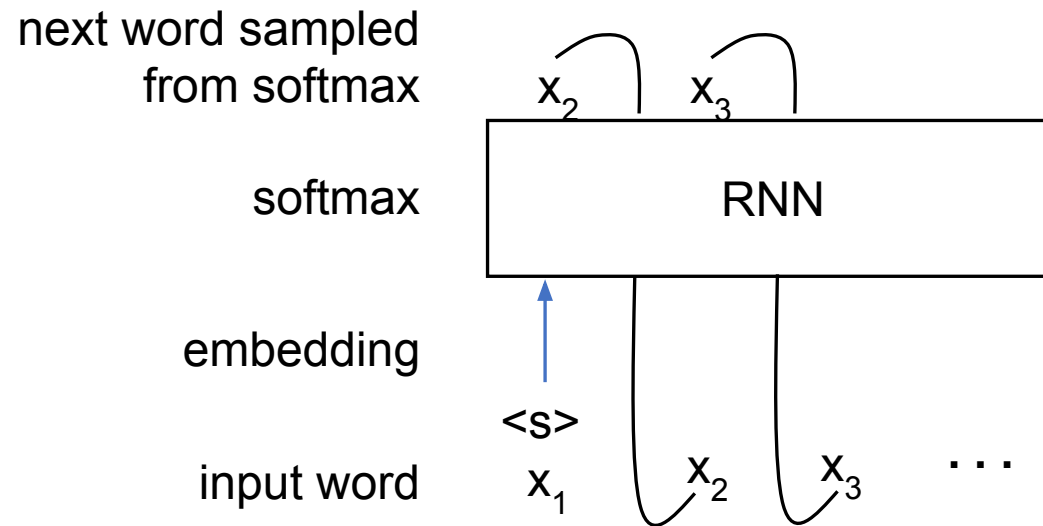


RNNs Can Also Be Stacked

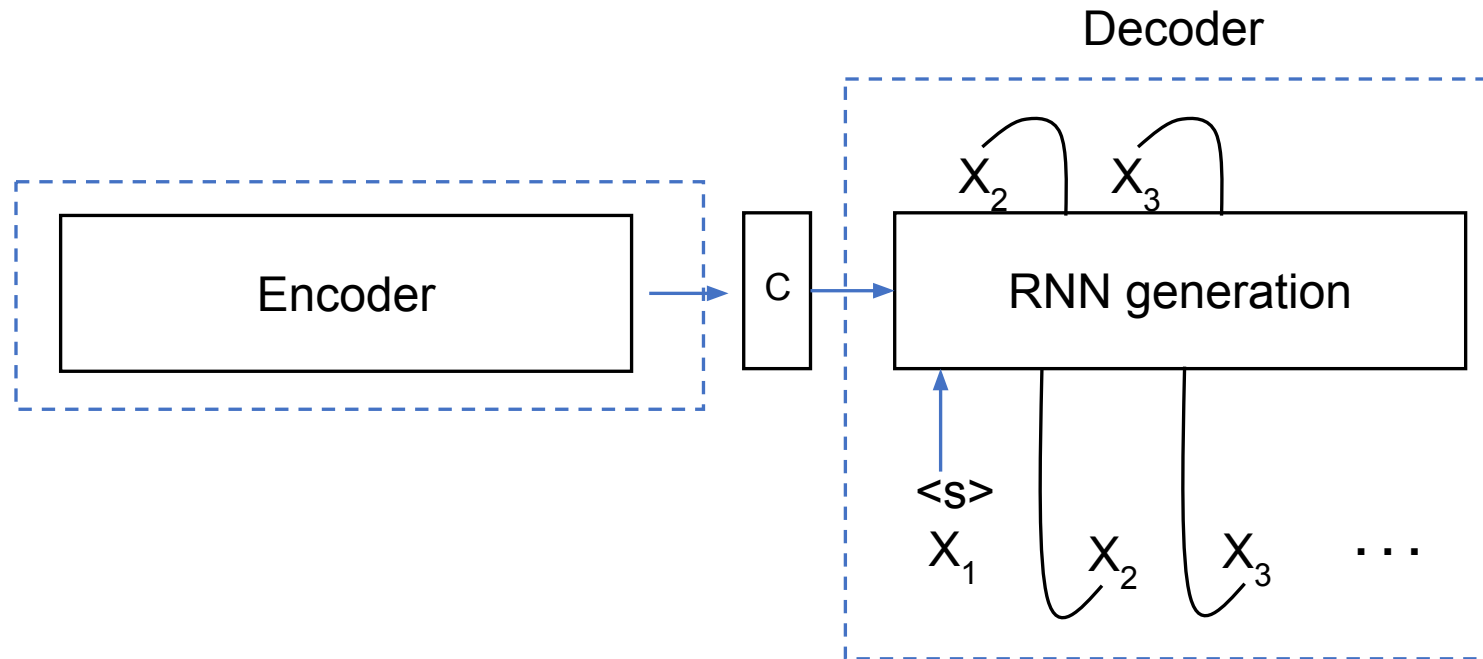


Each of these layers could also be a **bi-directional** RNN

Autoregressive Generation of Sequences



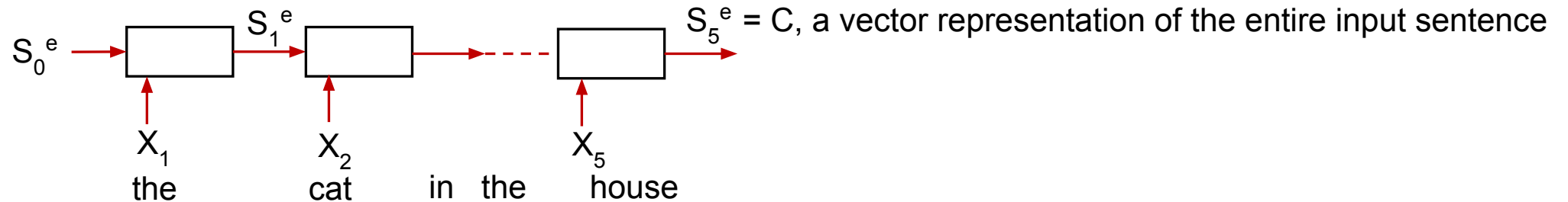
Generating From Context



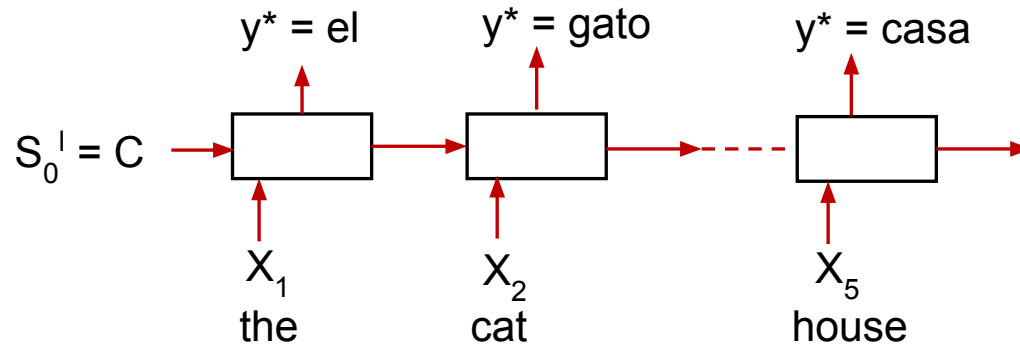
- C can represent an encoding of prior context using an RNN
- C can also represent encoding of anything else —e.g. an image, so that generation (the decoder) produces a caption

Encoder-Decoders for Machine Translation

Encoder

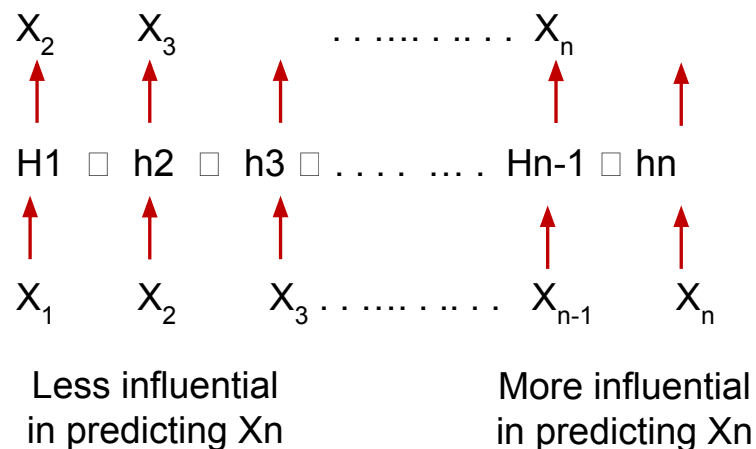


Decoder



Long Short-term Memory (LSTM) (1/4)

- A problem for the original RNN: earlier words have less influence than later words
- It is therefore more difficult to capture longer-distance dependencies



Long Short-term Memory (LSTM) (2/4)

The flights [the airline was cancelling after a serious storm knocked out power] were full

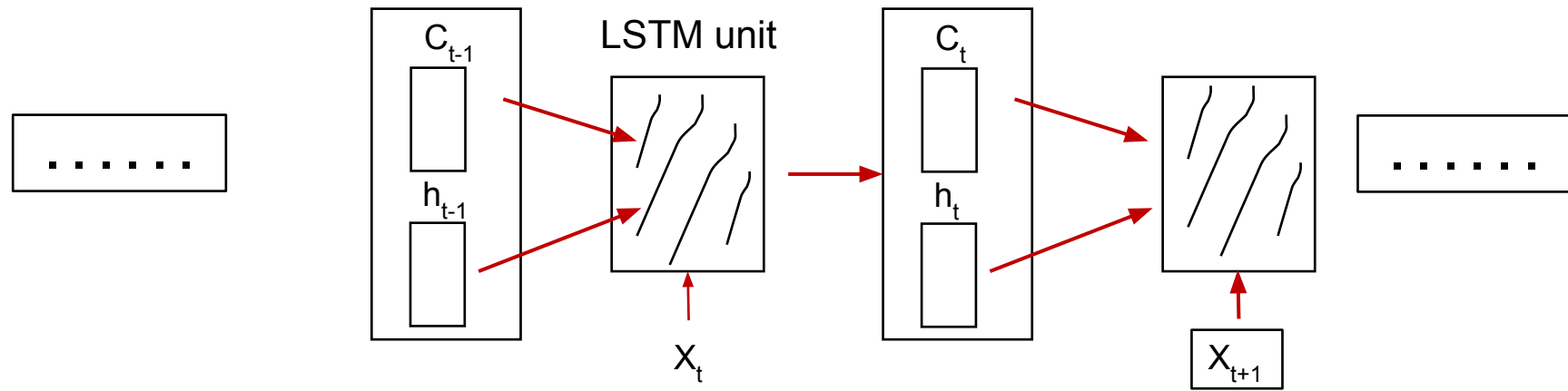
PL

PL

long-distance dependency



Long Short-term Memory (LSTM) (3/4)



- LSTM: uses “gates” to control propagation of information
- Same architecture as RNN, just different internal architecture for the units!

Long Short-term Memory (LSTM) (4/4)

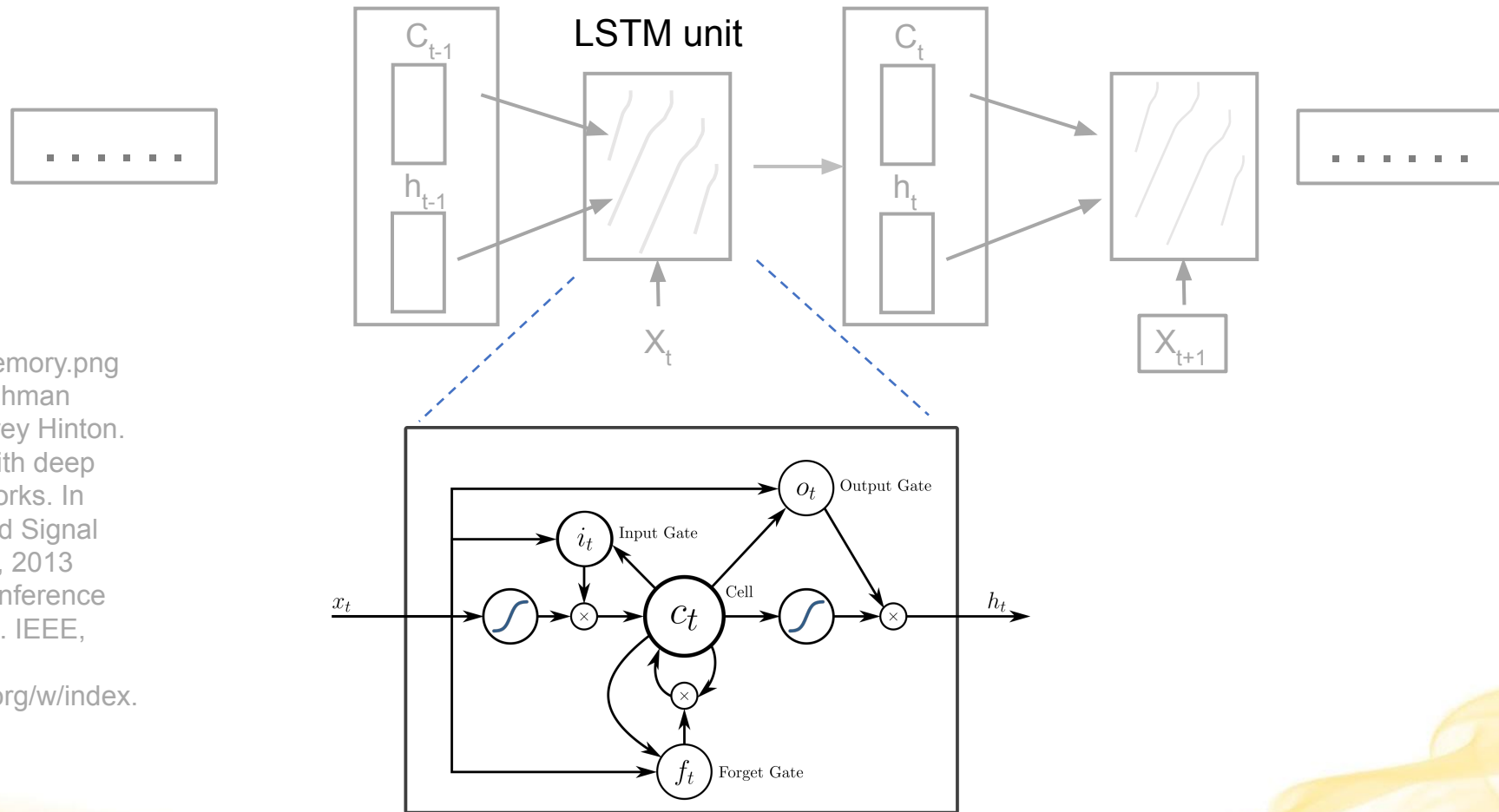
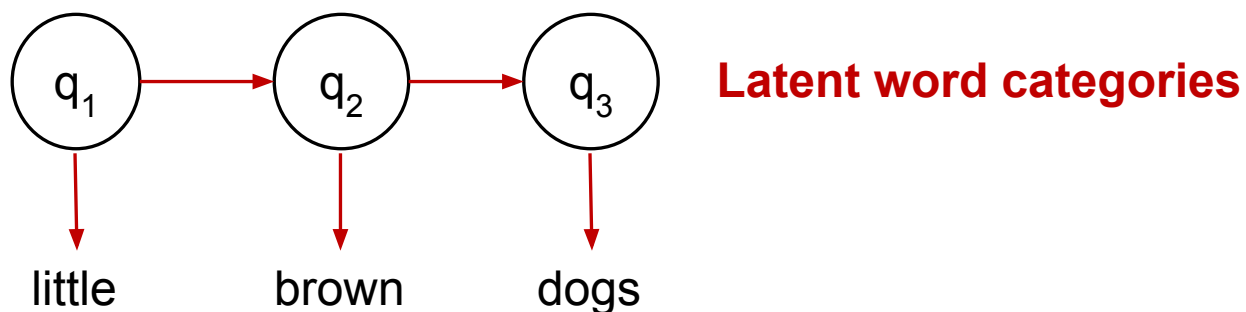


Image source:
Long_Short_Term_Memory.png
Alex Graves, Abdel-rahman
Mohamed, and Geoffrey Hinton.
Speech recognition with deep
recurrent neural networks. In
Acoustics, Speech and Signal
Processing (ICASSP), 2013
IEEE International Conference
on, pages 6645–6649. IEEE,
2013., CC BY-SA 4.0,
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php?curid=59931189](https://commons.wikimedia.org/w/index.php?curid=59931189)

Sequential Neural Models and Syntax (1/4)

- Recall that finite state automata:
 - Capture observable structure but do not represent latent/abstract categories
 - Do not capture important long-range linguistic dependencies
- HMMs learn latent structure,



Sequential Neural Models and Syntax (2/4)

- With fewer parameters than n-gram models, they are more effective at capturing generalizations.
 - N-gram model: $O(Vn)$ parameters over a vocabulary of size V
 - HMM with S states: $O(S^2) + O(V*S)$ parameters, where typically $S \ll V$
- But HMMs are still fundamentally finite-state and therefore fail to model long-distance dependencies

Sequential Neural Models and Syntax (3/4)

- In contrast, sequential neural models like RNNs and LSTMs are not strictly finite-state
 - Infinite number of states can be encoded by continuous-valued hidden state vectors.
 - Current state can depend on arbitrarily distant previous information
 - LSTMs further facilitate “remembering” relevant information for later in the string

Sequential Neural Models and Syntax (4/4)

- There is evidence that sequential neural models can capture long-distance dependencies and other aspects of hierarchical syntactic structure.

The keys the man that I saw lost { is
are } LSTM will correctly assign
“are” a much higher
probability

- Probes” like this, and other methods, suggest that deep neural models of language are implicitly representing structure

BERT: Bidirectional Encoder Representations from Transformers (1/2)

- Word-piece tokens
- Masked Language Models
 - The man went to the [MASK] to buy a [MASK] of milk.
 - Predicts from context both left-to-right, and right-to-left (hence, “bidirectional”)
 - Learning objectives: (1) predict masked word, (2) predict next sentence

BERT: Bidirectional Encoder Representations from Transformers (2/2)

- BERT representations are powerful and easily amenable to many fine-tuning tasks
 - But they are complex and expensive to train
- Multilingual BERT (m-BERT, Devlin et al. 2019)
 - Same architecture as BERT, training on a multilingual dataset
 - Uses a shared vocabulary of ~100 languages (single model)
 - Trained on Wikipedia pages across many different languages